

Loops in Java

A loop is execute a block of code repeatedly until a condition becomes false.

condition True

↓
Run code

↓
condition True

↓
run again

↓
condition True

↓
run again

↓
condition false

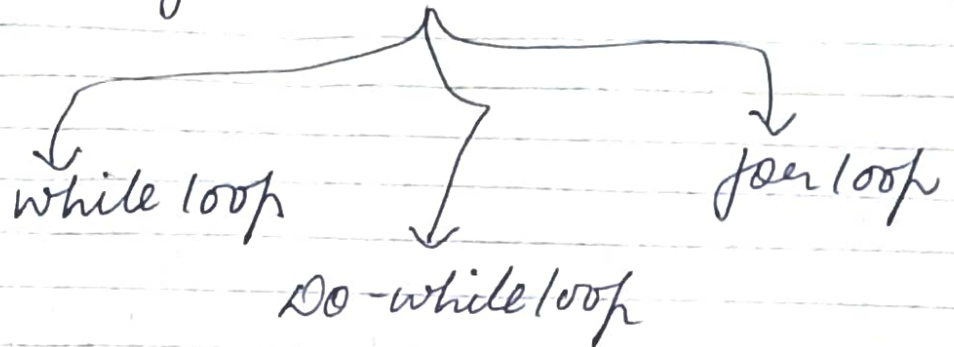
↓
stop

initialization

condition

statements

update



while loop (Best when iteration are unknown)

The while loop executes a block of code as long as the condition is true.

initialization;
while (condition)

{
statements;
update;
}

eg: int i = 1;

while (i <= 5)

{
s.o.p.ln ("Hello");

i++ }
update

// Hello
Hello
Hello
Hello
Hello

Also called entry controlled loop because condition is checked before the loop executes.

- Infinite loop is a loop whose condition never become false.

do-while loop (Exit-Controlled loop)

A do-while loop executes the block of code first and checks the condition afterward.

↓
The loop body always executes at least once

```
do {  
    // statements  
} while (condition);
```

do these statements
↓
Now check condition

↓
If true
↓
Repeat
↓
If false
↓
stop

eg:

```
int i = 1;  
do {  
    s.o.pln(i);  
    i++;
```

```
} while (i <= 5);
```

for loop (Best when iteration are known)

A for loop is used when we know approximately how many times we want to repeat a block of code.

```
for (initialization; condition; updation)  
{  
    statements;  
}
```

eg:

```
for (int i = 1; i <= 5; i++)  
{  
    s.o.pln(i);  
}
```

← one thing inside another thing

Nested loops in Java

Nested loop is a loop inside another loop.

- The outer loop controls rows.
- The inner loop controls columns.

```
for (initialization1; condition1; updation1)
{
    for (initialization2; condition2; updation2)
    {
        statements;
    }
}
```

eg:

```
for (int i = 1; i <= 3; i++) // repeat 3 rows
{
    for (int j = 1; j <= 5; j++)
    {
        // print 5 stars in each row
    }
}
```



```

eg: public class Main {
    psvm (String [] args) {
        for (int i=1; i<=3; i++) {
            for (int j=1; j<=5; j++) {
                S.O.P ( )
            }
        }
    }
}

```

output

```

  x x x x x
 x x x x x
 x x x x x

```

```

eg: Print no's
    1 1 1
    2 2 2
    3 3 3

```

```

for (int i=1; i<=3; i++)
{
    for (int j=1; j<=4; j++)
    {
        S.O.P (i);
    }
    S.O.P ( );
}

```